



“Gheorghe Asachi” Technical University of Iasi, Romania



IMPACT MECHANISM OF VEGETATION SLOPE SURFACE ON THE RUNOFF AND SEDIMENT PRODUCING PROCEDURE OF JINYANG ANCIENT CITY IN DIFFERENT PERIODS

Ruihua Shang^{1*}, Furong Sun², Hongxiong Lu², Pengju Han³, Jie Fu¹, Weiwei Liu⁴

¹College of Architecture, Taiyuan University of Technology, Taiyuan, Shanxi, China

²Jinci Museum, Taiyuan Cultural Relics Protection Research Institute, Taiyuan, Shanxi, China

³College of Civil Engineering, Taiyuan University of Technology, Taiyuan, Shanxi, China

⁴Taiyuan Institute of cultural relics and Archaeology, Taiyuan, Shanxi, China

Abstract

Primary vegetation covering was a rising technology for restoring surface diseases of earthen sites. However, the mechanism of herbs on the infiltration, runoff and sediment producing were unclear. In the in-situ artificial rainfall experiments in Jinyang Ancient City, the effects of different plant growth conditions in different periods on earthen sites slopes were compared with. The conclusions were as follows. The rainfall of 13.4mm was the threshold for changes in soil volume moisture content at depth in 0.5m. The fluctuation amplitude and the peak migration rate of soil moisture were positively correlated with soil porosity. The time of initial peak of soil surface volume moisture content was negatively correlated with its initial value before rainfall and rainfall intensity. The rainfall intercepted by plants was in a dynamic equilibrium state. Root splitting fissures and unstable water storage spaces not only had the function of water storage, but also had the one of reducing runoff velocity. For fresh herbs and withered herbs, interception yield before runoff producing Q_0 was negatively correlated with runoff starting time T_0 , and the interception effect of withered herbs was better than fresh ones. The impact of rainfall intensity on the percentage of runoff production C_{ro} of fresh herbs decreased over time, and the impact of rainfall intensity on the percentage of sediment production C_{sy} of fresh herbs increased over time. Fresh herbs and withered herbs enhanced the erosion resistance by intercepting and increasing the infiltration rate of rainfall and reducing the kinetic energy of rainwater. The stems and leaves blocked sunlight, regulating and creating a relatively stable microclimate environment to reduce evaporation of primary vegetation covering. It confirmed that fresh herbs and withered herbs had regulatory effects on conservation of earthen sites in Jinyang Ancient City. It would provide data support for the protection of earthen sites in large scales.

Key words: Jinyang Ancient City, primary vegetation, rainfall experiment, runoff producing, sediment producing

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1. Introduction

As one of the earliest utilized building materials, soil is closely related to human life. Earthen sites have witnessed the development of human civilization and have held historical, cultural, and artistic value. However, the increasing climate events of extreme prolonged heavy rainfall (Yang et al., 2019) accelerated the surface defects on earthen sites

(Dai et al., 2015; Wang and Yao, 2001; Wang et al., 2013; Zhou, 2011). The main manifestations of its damage are the volume reduction of the earthen sites, and skin-forming and fissure-forming on the surface of earthen sites. Moreover, if the earthen sites entered the northern freezing period prematurely without being completely dried, it would increase the risk of rainfall erosion, wind erosion, and freeze-thaw erosion (Shang et al., 2022a).

* Author to whom all correspondence should be addressed: e-mail: shangruihua@tyut.edu.cn; Phone: +8618535107598

To solve above problems, the primary vegetation covering has attracted much attention due to its advantages such as wide growth range, low cost, and excellent erosion and weathering resistance. Based on the vertical covering structure, a plant can be divided into two parts. They are surface stems and leaves, and underground roots. The surface stems and leaves can intercept rainfall, can weaken the kinetic energy of raindrops, and can increase infiltration (Adekalu et al., 2007; Han et al., 2021). The underground roots can anchor, reinforce, and bind the soil (Chen et al., 2008). Moreover, the fissures caused by root splitting can enhance the infiltration capacity of soil (Bi et al., 2020; Zhang, 2021).

It is known that the main factors affecting the processes of infiltration, runoff, and sediment production include rainfall intensity, slope length, primary vegetation cover, initial soil moisture content, and litter thickness (Chen et al., 2017; Christoph, 2017; Chu and Zhang, 2015; Si et al., 2013). Moreover, soil texture, bulk density, and soil stratification (Huang et al., 2013) also impact rainfall infiltration. Chang et al. (2019) confirmed that plant cover has a lag effect on rainfall and runoff production. Sun et al. (2017) proved that the reduction in runoff production due to litter is significant, and its runoff yield is consistently smaller than that from bare soil. It has been confirmed that regardless of the initial conditions, the sediment concentration in the later stage tends to stabilize (Liu et al., 2014; Lu et al., 2016). It has been verified that grass cover, species, and changes in herbs impact sediment production (Gan et al., 2010; Wu et al., 2010; Yang et al., 2016). Li et al. (2011) demonstrated that the sediment concentration measured on a 10m slope is similar to that on a 5m slope, as the kinetic energy of raindrops and overland flow energy is insufficient to support significant soil erosion. However, the aforementioned studies have primarily focused on indoor simulated

rainfall experiments. There is a lack of research on the processes of infiltration, runoff and sediment producing. Moreover, the vertical depth of infiltration research often remains within 1 m.

Based on the above shortcomings leading to knowledge gaps that require further verification, we hypothesize that there is no any relationship among the infiltration, runoff producing, and sediment producing on soil slope under different plant growth conditions in different times. This study utilizes the method of in-situ artificial rainfall simulation to analyze the effects of herbs on above processes. It aims to provide the references for promoting the reinforcement method of primary vegetation covering for earthen sites.

2. Experimental methodology

2.1. Experimental area

The western city wall of Jinyang Ancient City is located in Jinyuan District, southwest of Taiyuan City, Shanxi Province, China. It includes exposed parts above ground and buried parts underground. The existing exposed part has a trapezoidal profile with distinct rammed layers, with a total residual length of approximately 800 m. The height is 3-8 m. The width of the bottom is about 15 m, and that of the top is 4-10 m. The overall plant distribution on the city wall is uneven. The western side of the wall is mixed with shrubs and small trees, while the eastern side is predominantly covered with herbs and vines.

This experiment selected the middle position of the western city wall in Jinyang Ancient City (Fig. 1). The plants in the experimental slope section were all wild herbs and vines. Those plants had shallow root system, mostly ranging from 200 mm to 300 mm, with weak main roots and dense fibrous roots, forming root-soil composites.

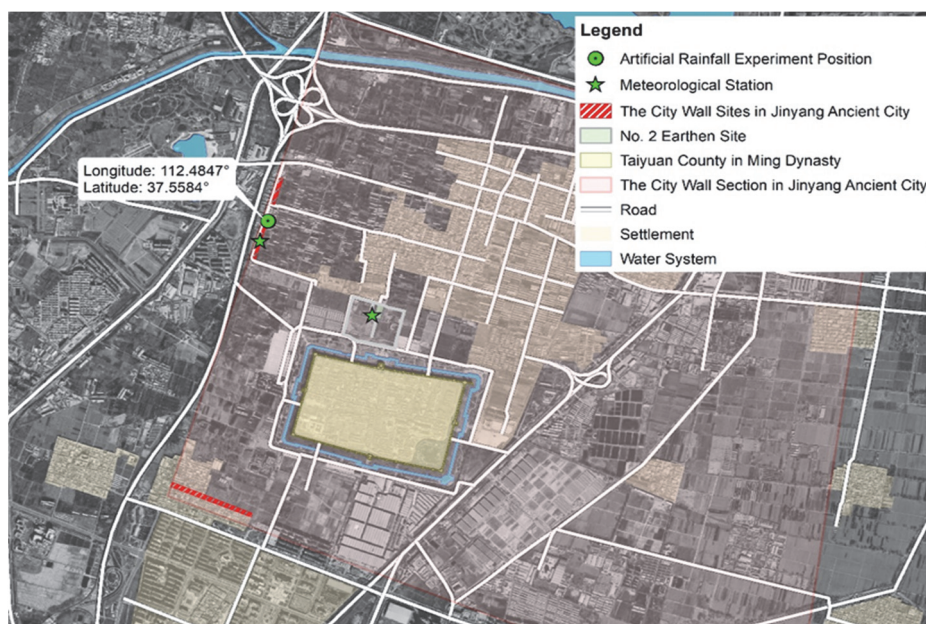


Fig. 1. Diagram of the location of meteorological station in Jinyang Ancient City

Soil tests conducted in Jinyang Ancient City (Shang et al., 2022a) showed that the predominant soil type was sandy loam, which was non-collapsible loess. As the depth of the soil increased, the soil moisture content also increased accordingly. The soil salinity was primarily composed of CaSO_4 and MgSO_4 , with pH values ranging from 7.1 to 7.45 and the soil specific gravity valued ranging from 2.69 to 2.7.

2.2. Experimental principles

The artificial rainfall experiment focuses on two issues. They are runoff producing and sediment producing. The sediment producing process is carried out simultaneously with the runoff producing process. And runoff producing is calculated based on the water balance equation (Eq. 1).

$$R = P - E - W_p - W_D - \Delta W_s - \Delta W_l \quad (1)$$

where: R is the runoff yield, P is the rainfall precipitation, E is the evapotranspiration, W_p is the plant interception, W_D is the water storage in different space, ΔW_s - increment of surface soil, ΔW_l - increment of internal soil, in mm.

Based on the short duration of the rainfall experiment, the proportion of evapotranspiration in the experimental process can be ignored. That means E can be taken as 0. Eq. (1) can be simplified as Eq. (2).

$$R = P - W_p - W_D - \Delta W_s - \Delta W_l \quad (2)$$

Sediment producing is calculated based on soil volume balance equation (Eq. 3).

$$\Delta V = V - V' \quad (3)$$

where: ΔV is the sediment producing volume, V is the experimental area volume of earthen sites before the rainfall, V' is the experimental area volume of earthen sites after the rainfall, in m^3 .

2.3. Experimental design

The experiments were multiple in-situ artificial rainfall ones conducted during a concentrated time period (Table 1). The experimental subjects were located at the same slope and position, facing east, with an angle of 37.5° . The experimental areas were set based on the number of water pipes (Fig. 2). It was known that one water pipe could cover an area of 1.75 m^2 . Scenarios 1-4 only used 2 water pipes, equivalent to 2.5 m^2 . Scenarios 5-7 used 7 water pipes, equivalent to 8.75 m^2 . As the season progressed, the primary vegetation covering on the slope varied slightly (Fig. 3), including Withered Herbs (abbreviated as WH), Fresh Herbs (abbreviated as FH), and Bare Soil (abbreviated as BS).

Firstly, the experiment data of the processes of infiltration, runoff producing, and sediment producing would be obtained. A steel frame structure was constructed to install artificial rainfall and monitoring equipment on the experimental slope. Artificial rainfall equipment included sprinkler and catchment device. A 4 cm-in-diameter water pipe was single-ended plugged and used for the sprinkler, with holes drilled at 20 mm intervals. Above multiple water pipes were placed along the slope, with 20 mm spacing between each pipe, extending from the top to the bottom of the slope. The water collection device needed to dig a small pit of $0.3 \text{ m} \times 0.3 \text{ m} \times 0.3 \text{ m}$ at the foot of the slope, with a built-in water bucket, and tilted the plastic cloth above the water bucket. It was required that the position of plastic cloth above the water bucket would had holes to collect rainwater. Based on the destructive effect of rainfall experiments in autumn and winter on earthen city walls, representative date in spring and summer were selected for concentrated experiments. It depended on the actual situation on site.

The monitoring parameters included precipitation, runoff yield, sediment yield, plant interception, and opening. The precipitation was recorded by using the difference in readings of impeller type water meter divided by experimental area. Runoff yield was obtained by measuring the volume of rainwater in the bucket with a measuring cylinder. Sediment yield was measured through the methods of sampling and drying weighing. Plant interception was calculated by subtracting the runoff and sediment production on slopes with plants from those without plants. The opening was qualitatively described through on-site observation.

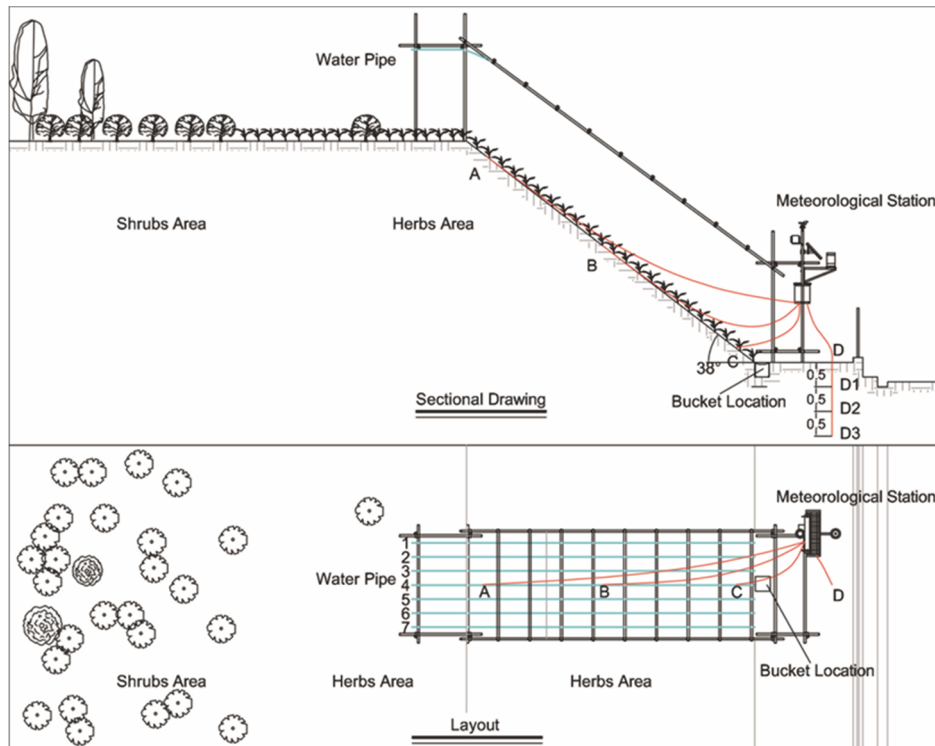
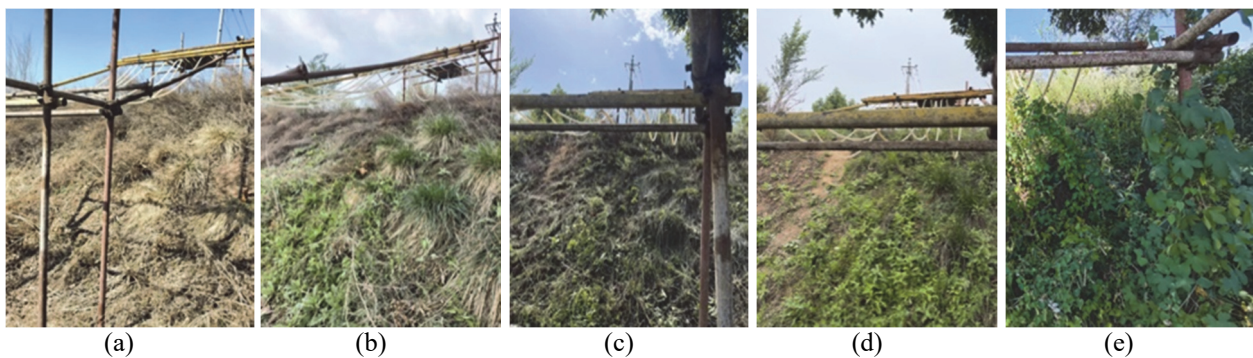
Next, it was necessary to supplement annual atmospheric precipitation monitoring data. A meteorological station was installed on the steel frame structure. The monitoring parameters included soil volume moisture content, precipitation, atmospheric temperature, and atmospheric humidity. S-RGB-M002 was selected as the rainfall gauge. The sensors of S-SMC-M005 were used to test the soil moisture at depths of 0.05 m, 0.5 m, 1 m and 1.5 m (Fig. 2). The sensor of S-THB-M002 RS3-B was selected to record air temperature and humidity. HOBO-U30-NRC meteorological station was used as the data logger. The monitoring time was recorded every 15 minutes.

2.4. Characterization parameters

This paper uses descriptive statistics to process and display data in the form of charts, and obtains data characteristics through summarizing and analyzing the data, with a greater focus on describing phenomena. Meanwhile, in response to the impact of rainfall on runoff and sediment producing, multiple comparative analysis methods are mainly used to compare the regulatory functions of bare soil, fresh herbs and withered herbs at different seasons.

Table 1. Statistics of relevant information for Scenarios 1-7

Scenarios		Experiments				Soil		Plant	
		Areas	Rainfall	Average rainfall intensity	Duration	Condition	Initial volume moisture content	Covering	Height
		m ²	mm	mm/min	hh:mm		m ³ /m ³		m
1	2022.3.27	2.5	266.8	2.448	1:49	Dry	0.129	65% WH + 5% FH + 30% BS	1.0~1.5
2	2022.8.14	2.5	611.08	3.796	2:41	Dry	0.133	100% FH	1.0~1.5
3	2023.3.27	2.5	306.68	0.757	6:45	Dry	0.132	65% WH + 5% FH + 30% BS	1.0~1.5
4	2023.4.25	2.5	628.96	1.782	5:53	Wet	0.279	60% WH + 40% FH	0.3~0.5
5	2023.6.1	8.75	338.66	1.75	3:13	Wet	0.285	40% WH + 60% FH	0.3~0.5
6	2023.6.26	8.75	200.66	0.667	5:01	Dry	0.145	5% WH + 95% FH	0.3~0.5
7	2023.8.22	8.75	338	0.901	6:15	Dry	0.134	100% FH	1.0~1.5

**Fig. 2.** Experimental plot of the western city wall in Jinyang Ancient City**Fig. 3.** Distribution of vegetation covering on slopes of Jinyang Ancient City in different months
(a) Scenario 3, (b) Scenario 4, (c) Scenario 5, (d) Scenario 6, (e) Scenario 7

As for the accuracy testing, just like quality checks for data accuracy, handling of missing data, and validation of sensor measurements, it still needs to be completed through future laboratory experiments. Runoff coefficient refers to the runoff yield generated by a unit of rainfall (Liu et al., 2014), which is numerically equal to the value of the cumulative runoff yield divided by rainfall during that time period. It could be defined by the ratio (Eq. 4). And other relevant parameters could be calculated by from Eq. (5) to Eq. (9).

$$C_{ro} = \sum_{i=1}^T R_i / \sum_{i=1}^T P_i \times 100\% \quad (4)$$

$$P = \frac{V_{sb}}{S} \times 1000 \quad (5)$$

$$I = \frac{P}{T} \quad (6)$$

$$R = \frac{V_{st}}{S} \quad (7)$$

$$I_{ro} = \frac{R}{T_{ro}} \quad (8)$$

$$Q_0 = \sum_{i=1}^{T_0} P_i \quad (9)$$

where: C_{ro} is the percentage of runoff production at different times, in %; R_i is the runoff yield at i minute, P_i is the precipitation at i minute, R is the runoff yield, P is the precipitation, Q_0 is the interception yield before runoff producing, in mm; V_{sb} is the total amount of precipitation, in m^3 ; V_{st} is the volume of runoff collected in the bucket, in L; S is the experimental area, in m^2 ; I is the rainfall intensity, I_{ro} is the runoff intensity, in mm/min; T is the rainfall time, T_{ro} is the runoff time, T_0 is the runoff starting time, in min.

Sediment yield coefficient refers to the sediment yield generated by a unit of rainfall (Liu et al., 2014), which is numerically equal to the value of the cumulative sediment yield divided by rainfall during that time period (Eq. 10). And other relevant parameters could be calculated by Eqs. (11-12).

$$C_{sy} = \sum_{i=1}^T Y_i / \sum_{i=1}^T P_i \quad (10)$$

$$Y = \frac{W_{sg} \times V_{st}}{V_{cy} \times S} \quad (11)$$

$$I_{sy} = \frac{Y}{T_{sy}} \quad (12)$$

where: C_{sy} is the percentage of sediment production at different times, in g/L; Y_i is the sediment yield at i minute, Y is the sediment yield, in g/ m^2 ; W_{sg} is the weight of sand, in g; P_i is the precipitation at i minute, V_{cy} is the volume of sampled runoff, in L; I_{sy} is the sediment intensity, in g/(min· m^2); T_{sy} is the sediment time, in min.

3. Results and discussion

3.1. Results

(1) Infiltration

The fluctuation study of soil moisture at different depths is an important indicator for studying the impact of precipitation on the infiltration of earthen sites. Increment of internal soil (ΔW_I) and increment of surface soil (ΔW_S) can be characterized by parameters such as soil volume moisture content at different depths, in m^3/m^3 .

(i) Moisture in internal soil

By comparing and analyzing the relationship between rainfall and soil volume moisture content at depths in 0.5 m, 1 m, and 1.5 m on a 0° slope, the results showed as follows.

(a) The rainfall of 13.4 mm (Fig. 4a) was the threshold for changes in soil volume moisture content at depth in 0.5 m. When cumulative rainfall was no bigger than 9.6 mm, the soil volume moisture contents at different depths remained unchanged. When cumulative rainfall was no less than 13.4 mm, the soil volume moisture contents at different depths increased with the increase of rainfall.

(b) The experiment showed a common sense outcome that the volatility of soil moisture followed a trend of $0.5\text{ m} > 1\text{ m} > 1.5\text{ m}$, when cumulative rainfall was no less than 13.4 mm. As shown in Table 2, when cumulative rainfall was 13.4mm, the soil volume moisture content at depth in 0.5 m increased, but the ones at depth in 1.0 m and 1.5 m had little changes. When cumulative rainfall was greater than 13.4 mm, all soil volume moisture contents at different depths increased with the increase of rainfall. That meant as the rainfall increased, the difference in soil volume moisture content before and after rainfall at different depths gradually increased. The fluctuation amplitude of soil volume moisture content was inversely proportional to the difference at depths. The shallower the depth was, the greater the fluctuation in soil volume moisture content was.

(c) After rainfall, the high humidity of soil needed to evaporate water to atmosphere, which had the characteristic of rapid evaporation. A few days later, the evaporation entered into a slow stage. But from an annual distribution perspective (Fig. 4), it was still steadily evaporating into the atmosphere. According to Table 3, the soil at depths in 0.5 m and 1.5 m were in a rapid evaporation stage within 1.5 hours, and it entered into a relatively stable evaporation stage after 1.5 hours. The soil at depth in 1 m was basically in a stable evaporation stage, with a change rate of $0.0019\text{ m}^3/(\text{m}^3 \cdot \text{h})$.

(d) The experiment showed a common sense outcome that the fluctuation of soil moisture was positively correlated with the fluctuation of annual temperature. Absolute air humidity was positively correlated with daily cumulative rainfall (Fig. 5). From March to November, the soil volume moisture content at different depths followed a rule of $0.5\text{ m} > 1.5\text{ m} > 1\text{ m}$. From late December to early February, the soil volume moisture content followed the trend of $1.5\text{ m} > 0.5\text{ m} > 1\text{ m}$.

(ii) Moisture in surface soil

Based on the principle of minimizing interference with the earthen city walls, it was not possible to research on soil volume moisture content at different depths, for the western city wall of Jinyang Ancient City was one of the fifth batch of National Major Cultural Relics Protection Units. What we could do was to monitor and analyze the relationship between rainfall and the soil volume moisture content

at depth in 0.05m on *A*, *B*, and *C* positions in Scenario 5 and the ones on *A* position in Scenario 7 (Fig. 6), which were located on a 37.5° slope. The results showed as follows.

(a) In Scenario 5, the first rainfall occurred from 0:20 to 2:45 and the second one occurred from 6:10 to 7:20. The soil surface volume moisture content at depth in 0.05m showed an upward trend.

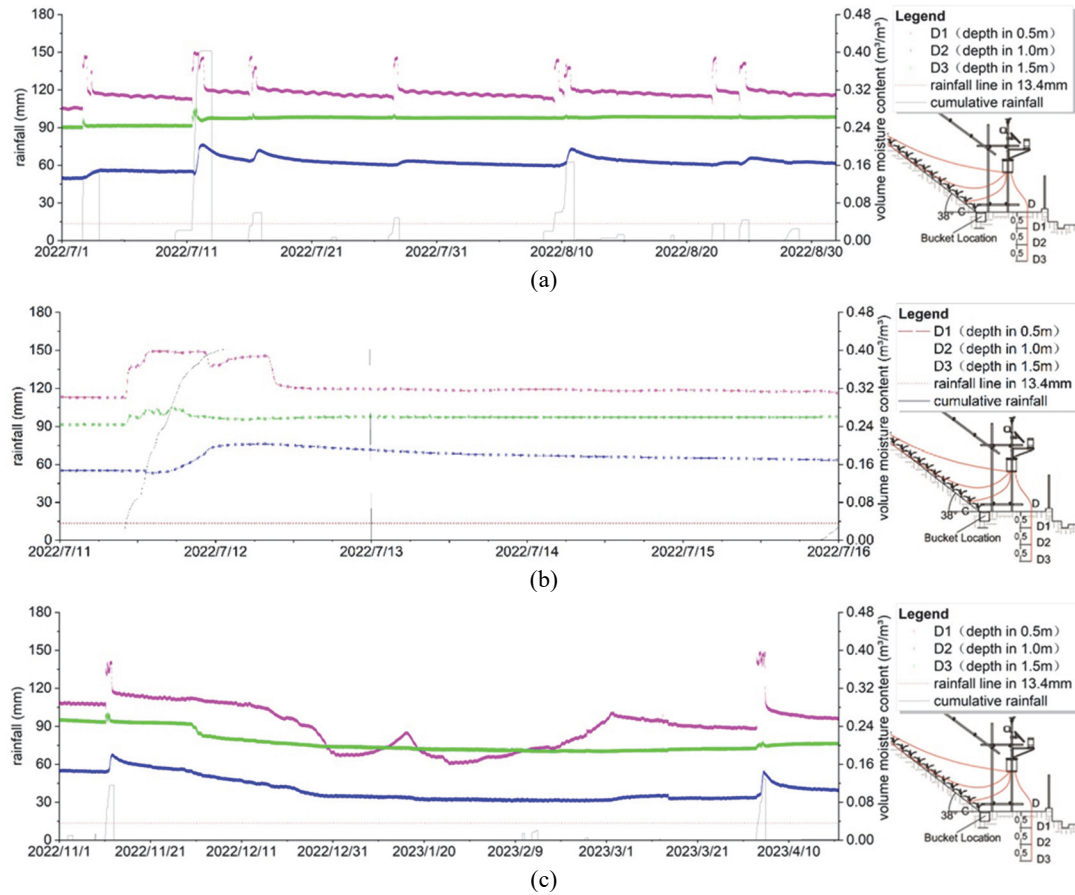


Fig. 4. Relationship between cumulative rainfall and soil volume moisture content at different depths, (a) rainfall line in 13.4mm, (b) gradual decrease in soil volume moisture content, (c) cold weather

Table 2. Statistics of soil volume moisture content during rainfall

Date	Rainfall (mm)	Soil volume moisture content (m^3/m^3)								
		<i>D</i> ₁			<i>D</i> ₂			<i>D</i> ₃		
		Before	Stop	Difference	Before	Stop	Difference	Before	Stop	Difference
8.28	9.6	0.3126	0.3119	-0.0007	0.1652	0.1666	0.0014	0.2625	0.2625	0
8.22	13.4	0.3068	0.3881	0.0813	0.1615	0.1659	0.0044	0.2618	0.2618	0
7.16	16.2	0.313	0.3765	0.0635	0.1623	0.1753	0.013	0.2618	0.2676	0.0058
7.12	151	0.301	0.3969	0.0959	0.1448	0.2037	0.0589	0.2443	0.2821	0.0378

Table 3. Statistics of gradual reduction of soil volume moisture content

Position	Date	Stop		Rapid dehumidification			Stable dehumidification		
		time	Soil volume moisture content	time	Soil volume moisture content	Change rate	time	Soil volume moisture content	Change rate
		hh:mm	m^3/m^3	hh:mm	m^3/m^3	$\text{m}^3/(\text{m}^3 \cdot \text{h})$	hh:mm	m^3/m^3	$\text{m}^3/(\text{m}^3 \cdot \text{h})$
D ₁	2022.7.12	7:45	0.3881	9:15	0.3308	0.0382	23:45	0.3191	0.0008
D ₂	2022.7.12	5:00	0.2259	—	—	—	23:45	0.1906	0.0019
D ₃	2022.7.11	18:30	0.2785	20:00	0.2639	0.0097	23:45	0.2581	0.0015

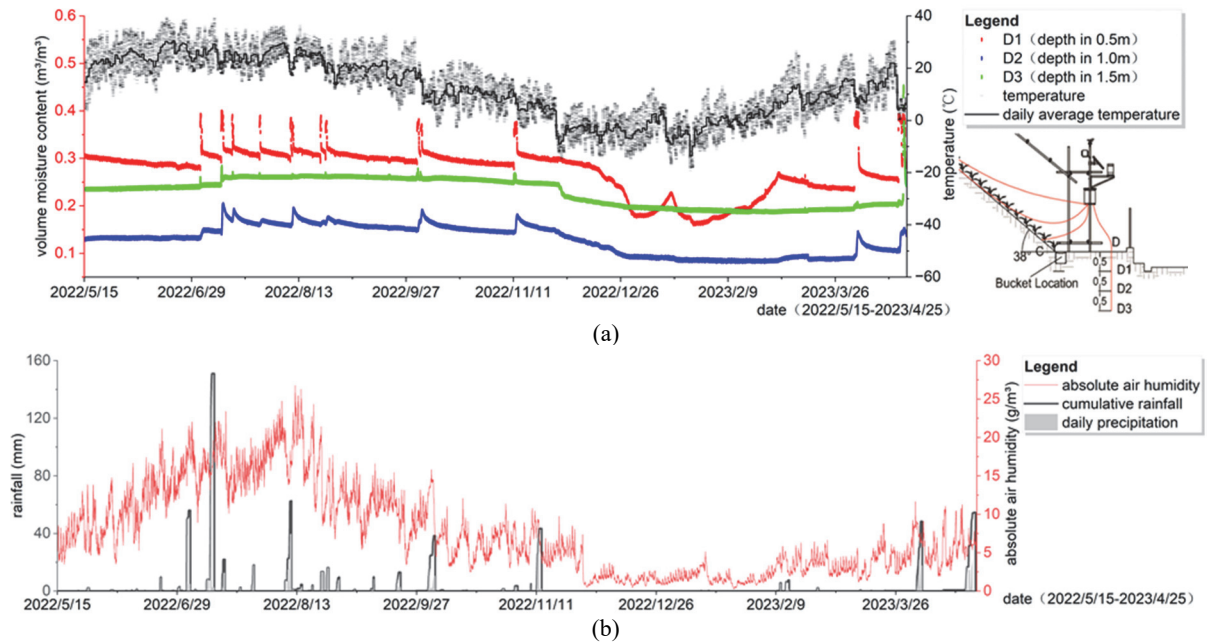


Fig. 5. Relationships among temperature, humidity, soil volume moisture content, and daily precipitation at different depths, (a) Temperature and soil volume moisture content, (b) Absolute air humidity and daily precipitation

Before rainfall, it followed the trend of $B > A > C$. During the first rainfall, C , A , and B reached their peak around 20 minutes, 50 minutes, and 80 minutes, respectively. In addition, the soil surface volume moisture content at depth in 0.05m followed the trend of $C > A > B$. During the intermittent rainfall, A , B , and C all showed a slightly downward trend, following the chronological order of A , C , and B . The decrease rate followed the trend of $C > A > B$. C reached its peak after 10 minutes of the second rainfall. A , and B reached their peak after 20 minutes. And the soil surface volume moisture content at depth in 0.05m followed the trend of $B > C > A$.

(b) In Scenario 7, the first rainfall occurred from 0:00 to 4:00 and the second one occurred from 5:30~6:20. During the first rainfall, the soil surface volume moisture content at depth in 0.05m showed an upward trend. A reached its peak around 30 minutes. During the intermittent rainfall, A showed a downward trend. A reached its peak after 15 minutes of the second rainfall.

(c) Comparing the relationship between rainfall intensity and the soil surface volume moisture content in Scenario 5 and Scenario 7, the results were as follows. The time of initial peak of soil surface volume moisture content was negatively correlated with its initial value before rainfall and rainfall intensity. The larger the initial value of soil surface volume moisture content was, and the higher the rainfall intensity was, the longer the initial peak time of soil surface volume moisture content was. Scenario 5 experienced sustained heavy rainfall before the artificial rainfall experiment, and the initial volume moisture content at depth in 0.05 m was $0.285 \text{ m}^3/\text{m}^3$. During the first rainfall, its average rainfall intensity was 1.645, with a peak occurring around 50 minutes. Scenario 7 had no rainfall before, and the initial volume moisture

content at depth in 0.05m was $0.134 \text{ m}^3/\text{m}^3$. And the average rainfall intensity was 0.864, with a peak occurring around 30 minutes. During the intermittent rainfall and the second rainfall, the soil surface volume moisture content at depth in 0.05m in both Scenario 5 and Scenario 7 showed a downward trend firstly and then an upward trend. Both the decrease rate and growth rate followed the trend of Scenario 7 > Scenario 5.

(2) Plant Interception and Openings

(i) Plant interception

Plant interception refers to the rainfall intercepted by plant branches or leaves before it falls to the ground. The process of plant interception lasts throughout the entire rainfall period. Through on-site experimental observation, the interception process had the following characteristics.

(a) At the beginning of rainfall, discrete water droplets were intercepted on the surface of branches, fresh leaves, and withered leaves. They were accumulated and then gathered together to form a water covering layer on branches and leaves. Gradually, the layer became thick. During this process, all rainwater was intercepted on the surface of the branches and leaves, and the interception yield had no relationship with rainfall intensity.

(b) As the rainfall continued, the gravity value of the water layer gradually increased and approached the adhesion value between branches and water. When plants reached the maximum interception capacity, these overweight droplets, stored on branches and leaves originally, were constantly replaced by new ones. They either fell into the ground through the gaps between branches and leaves to form a through forest rain, or flowed along the branches to the ground.

(ii) Water storage from plant openings

There were no obvious depressions on the experimental slope, so only plant openings were considered. Observation showed that the plant openings included two factors. One was the root splitting fissures formed during the growth of plant roots, which formed a water storage space and increased the channel for rainwater infiltration. The other one was the space of unstable water storage. It came from clustered plants. Due to their growth habits of clustering, the slender leaves tend to accumulate withered herbs, constructing unstable water storage spaces. Both of them increased the water content into the earthen sites. During the runoff process, rainwater gradually filled the small openings. After that, these openings had overflow phenomena. Research showed

that the velocity of overflow was slightly lower than that of runoff.

(3) Rainfall and sediment producing

(i) Runoff starting time T_0 and interception yield before runoff producing Q_0

Seasonal changes had an impact on T_0 and Q_0 . It was due to the different distribution of vegetation coverage on the underlying surface of earthen sites in different seasons. Table 1 showed that scenarios 1 and 2 had similar average rainfall intensity and similar initial soil volume moisture contents, so did scenarios 3 and 7. However, as shown in Table 4, compared to scenarios 2 and 7, the scenarios 1 and 3 had a delay of 3 minutes and 6 minutes on T_0 , and they increased by 7.92 L/m² and 5.15 L/m² on Q_0 .

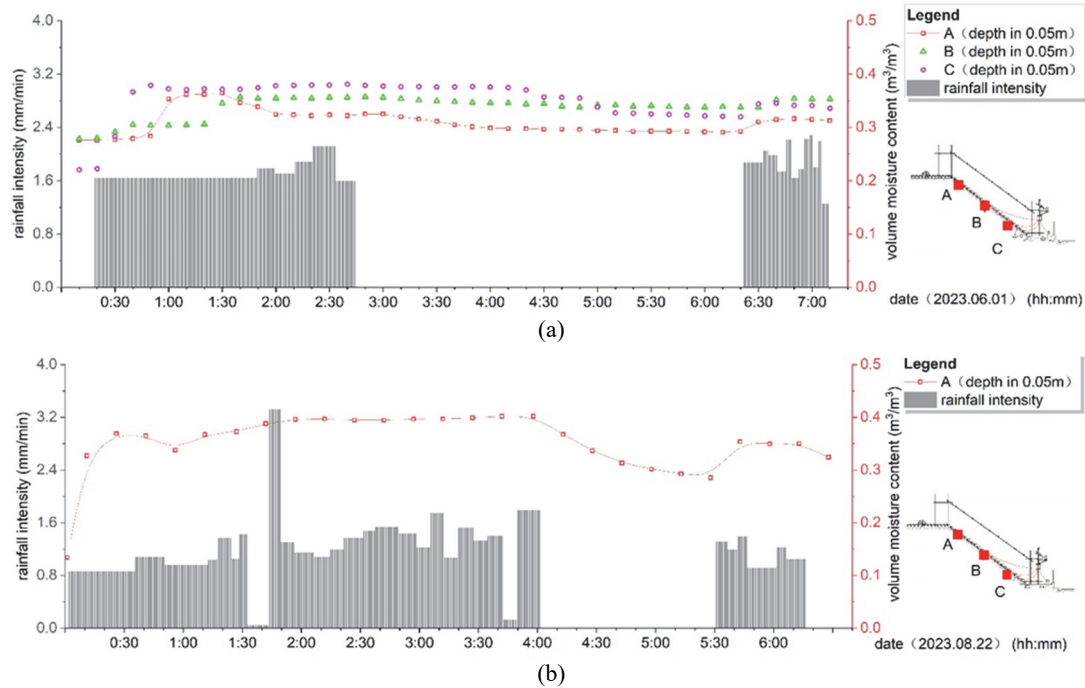


Fig. 6. Relationship between rainfall intensity and soil volume moisture content, (a) Scenario 5, (b) Scenario 7

Table 4. Statistical table of runoff starting time

Seasons	Scenarios	T_0	$\bar{I}_{rf}$	Q_0
		min	mm/min	mm
Spring	1	8	2.5	20
	3	13	0.862	11.2
Summer	2	5	2.416	12.08
	7	7	0.864	6.05
	6	12	1.351	16.22

(a) Comparing scenarios 1, 2, 3, 6, and 7, T_0 followed a rule of Scenario 3 > Scenario 6 > Scenario 1 > Scenario 7 > Scenario 2.

Rainfall intensity in scenarios 3 and 7 was less than 1, and T_0 followed a rule of Scenario 3 ($\bar{I}_{min}$) > Scenario 7 ($\bar{I}_{min}$), which was 65% WH + 30% BS ($\bar{I}_{min}$) > 95% FH ($\bar{I}_{min}$).

In scenario 2 and 7, the plant distribution was 100% Fresh Herbs, scenario 7 ($\bar{I}_{min}$) > Scenario 2

($\bar{I}_{max}$), which was FH ($\bar{I}_{min}$) > FH ($\bar{I}_{max}$). In other words, the higher the rainfall intensity was, the higher value in T_0 the fresh herbs had. It was concluded that FH ($\bar{I}_{min}$) > FH ($\bar{I}_{mid}$) > FH ($\bar{I}_{max}$).

(b) The interception before runoff producing Q_0 consisted of two parts- plant interception and soil infiltration. Comparing scenarios 1, 2, 3, 6, and 7, Q_0 followed a rule of Scenario 1 > Scenario 6 > Scenario 2 > Scenario 3 > Scenario 7.

The Q_0 of scenarios 2 and 7 followed a rule of Scenario 2 ($\bar{I}_{max}$) > Scenario 7 ($\bar{I}_{min}$), which was FH ($\bar{I}_{max}$) > FH ($\bar{I}_{min}$). It indicated that the greater the rainfall intensity was, the greater value in Q_0 of fresh herbs had. That was FH ($\bar{I}_{max}$) > FH ($\bar{I}_{mid}$) > FH ($\bar{I}_{min}$). Based on the above conclusion and Scenario 6 ($\bar{I}_{min}$) > Scenario 2 ($\bar{I}_{max}$), it was clear that WH ($\bar{I}_{mid}$) > FH ($\bar{I}_{max}$).

Based on the above conclusions, it can be concluded that Q_0 followed the rules of WH ($\bar{I}_{mid}$) > FH ($\bar{I}_{max}$) > FH ($\bar{I}_{mid}$) > FH ($\bar{I}_{min}$). It indicated that for fresh herbs and withered herbs, the higher the rainfall intensity was, the higher value the interception before runoff producing had, though the runoff starting time was smaller. Q_0 was negatively correlated with T_0 . Moreover, the interception effect of withered herbs was better than fresh herbs.

(ii) Runoff coefficient C_{ro} and sediment yield coefficient C_{sy}

According to Eq. (4), the higher the value of C_{ro} was, the better the runoff producing effect of the underlying surface (Fig. 7a) was. According to Eq. (5), the higher the value of C_{sy} was, the better the sediment producing effect of the underlying surface (Fig. 7b) was. According to Table 5, results showed as follows.

(a) The impact of rainfall time on the values of C_{ro} and C_{sy} decreased over time. The values of C_{ro} and C_{sy} before T_0 were 0 at the beginning, which meant the interception effect of the underlying surface before T_0 was the strongest, reaching 100%. As time went on, the values of C_{ro} and C_{sy} rapidly increased, followed by a slow growth rate until they stabilized. In other words, the interception effect decreased over time.

(b) Comparing scenarios 2, 6, and 7, when rainfall occurred within 60 minutes, the values of C_{ro} followed a trend of Scenario 6 > Scenario 2 > Scenario 7. When the rainfall occurred within 120 or 180 minutes, the values of C_{ro} followed a trend of Scenario 7 > Scenario 2. The C_{ro-60} of scenarios 2 and 7

followed a rule of Scenario 2 ($\bar{I}_{max}$) > Scenario 7 ($\bar{I}_{min}$), which was FH ($\bar{I}_{max}$) > FH ($\bar{I}_{min}$). In other words, the larger the rainfall intensity was, the larger value the C_{ro-60} for fresh herbs had. That was, FH ($\bar{I}_{max}$) > FH ($\bar{I}_{mid}$) > FH ($\bar{I}_{min}$). Based on the above conclusion and Scenario 6 ($\bar{I}_{mid}$) > Scenario 2 ($\bar{I}_{max}$), it can be concluded that WH ($\bar{I}_{mid}$) > FH ($\bar{I}_{max}$).

The C_{ro-120} and C_{ro-180} of scenarios 2 and 7 followed a rule of Scenario 7 > Scenario 2, which was FH ($\bar{I}_{min}$) > FH ($\bar{I}_{max}$). In other words, the greater the rainfall intensity was, the smaller value of C_{ro-120} and C_{ro-180} the new herb had. That was, FH ($\bar{I}_{min}$) > FH ($\bar{I}_{mid}$) > FH ($\bar{I}_{max}$).

That meant, as time went by, the values of C_{ro} changed gradually. There was a positive correlation between C_{ro-60} of fresh herbs and rainfall intensity, while there was a negative correlation between C_{ro-120} and C_{ro-180} of fresh herbs and rainfall intensity.

(c) Comparing scenarios 2, 6, and 7, when rainfall occurred within 60 or 180 minutes, the values of C_{sy} followed a rule of Scenario 2 > Scenario 6 > Scenario 7. When rainfall occurred within 120 minutes, the values of C_{sy} followed a rule of Scenario 6 > Scenario 2 > Scenario 7.

The distribution of plants in scenarios 2 and 7 was 100% Fresh Herbs. The values of C_{sy-60} , C_{sy-120} , and C_{sy-180} followed a rule of Scenario 2 ($\bar{I}_{max}$) > Scenario 7 ($\bar{I}_{min}$), which was FH ($\bar{I}_{max}$) > FH ($\bar{I}_{min}$). In other words, the greater the rainfall intensity was, the larger values of C_{sy-60} , C_{sy-120} , and C_{sy-180} the fresh herbs had. That was FH ($\bar{I}_{max}$) > FH ($\bar{I}_{mid}$) > FH ($\bar{I}_{min}$). That meant, as time passed, the values of C_{sy} followed a rule of FH ($\bar{I}_{max}$) > FH ($\bar{I}_{mid}$) > FH ($\bar{I}_{min}$). The greater the rainfall intensity was, the greater value the C_{sy} of fresh herbs had. In other words, there was a positive correlation between C_{sy} of new herbs and rainfall intensity. Based on the value of C_{sy-120} was satisfied with Scenario 6 ($\bar{I}_{mid}$) > Scenario 2 ($\bar{I}_{max}$), it induced that WH ($\bar{I}_{mid}$) > FH ($\bar{I}_{max}$)

Table 5. Statistical Table of slope runoff coefficient

Scenarios			T	$\bar{I}_{rf}$	P_{ac}	R_{ac}	C_{ro}	Y	C_{sy}
			min	mm/min	mm	mm	%	g/m^2	g/L
Spring	March	1	40	2.46	98.4	0.484	0.492	31.15	0.317
			109	2.448	266.8	5.528	2.072	107.575	0.403
		3	48	0.862	41.36	0.21	0.508	0.808	0.02
			117	0.891	104.24	0.571	0.548	1.436	0.014
			172	0.968	166.48	2.499	1.501	23.185	0.139
			405	0.757	306.68	11.219	3.658	68.222	0.222
	April	4	353	1.782	628.96	52.036	8.273	—	—
Summer	June	5	193	1.75	338.66	9.95	2.938	—	—
		6	53	1.05	55.62	1.109	1.993	3.799	0.068
			112	1.052	117.78	5.590	4.677	22.213	0.189
			173	0.852	147.45	10.583	7.177	37.216	0.252
			301	0.667	200.66	25.109	12.513	56.589	0.282
	August	2	43	2.363	101.6	0.458	0.451	7.992	0.079
			118	2.488	293.6	7.634	2.6	27.846	0.095
			161	3.796	611.08	24.938	4.081	181.311	0.297
		7	49	0.931	45.63	0.617	0.014	0.683	0.015
			115	1	121.02	6.171	5.1	5.869	0.049
			177	1.115	203.38	16.194	7.962	14.929	0.073
			375	0.901	338	33.69	9.968	23.455	0.069

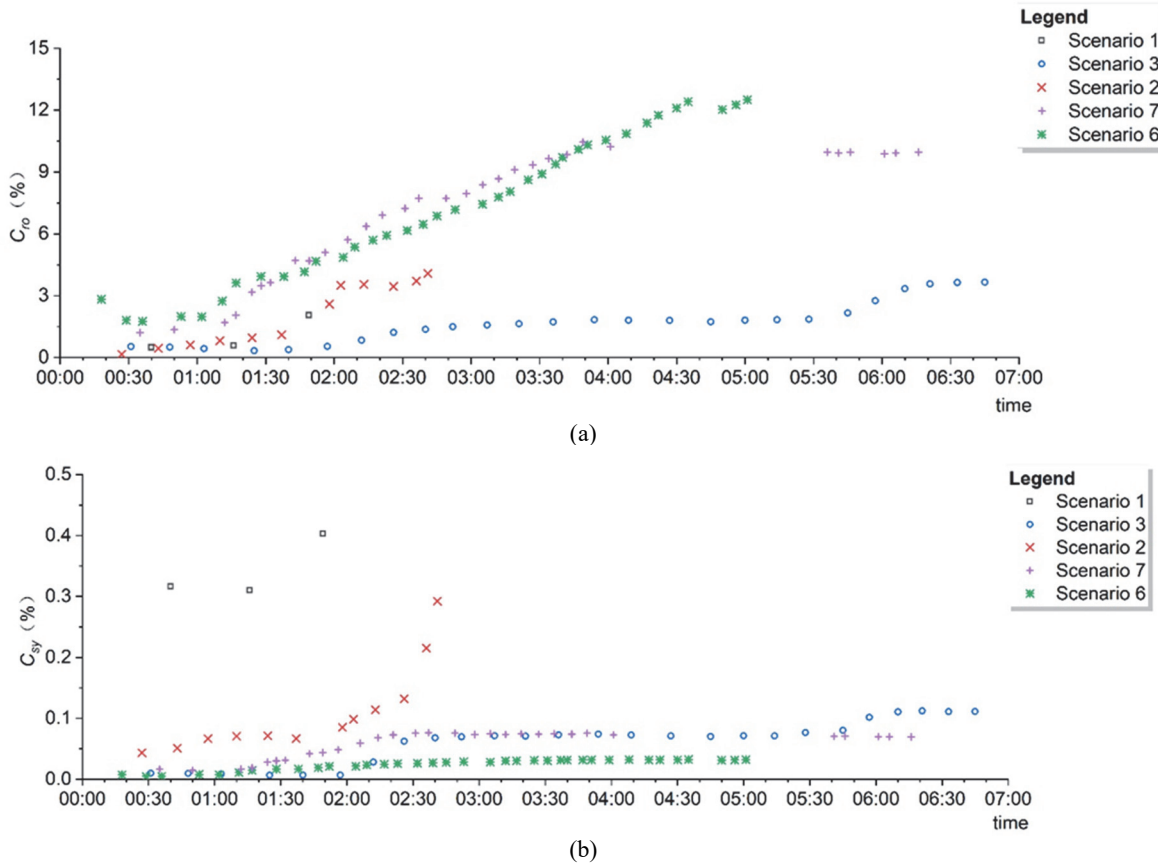


Fig. 7. Distribution of runoff coefficient and sediment yield coefficient, (a) Runoff coefficient, (b) Sediment yield coefficient

3.2. Discussion

(1) Resistant mechanism of primary vegetation covering to rain erosion

Fresh herbs and withered herbs enhanced the erosion resistance by intercepting and increasing the infiltration rate of rainfall, as well as reducing the kinetic energy of rainwater. They absorbed rainwater to complete the plant interception process. They combined their rain resistance and water storage functions to increase the infiltration rate of rainfall. They used their own elastic deformation to reduce the kinetic energy of rainwater.

(i) Infiltration at different depths

The reasons why descent values of soil volume moisture content at different depths decreased over time was as follows. Due to the soil at depth in 0.5 m being close to the soil surface, its evapotranspiration process occurred rapidly. That was why it increased first, and then decelerated. The fluctuation value of soil volume moisture content at depth in 1.5 m was the smallest one among the ones at depth in 0.5 m, in 1.0 m and in 1.5 m. Therefore, it took least time to achieve water dynamic equilibrium. It was consistent with the research findings from Yang and Sui (1997).

Shang et al. (2022b) pointed out that the distribution of soil porosity at different depths followed the rule of 0.5 m > 1 m > 1.5 m. The above rule order was consistent with the fluctuation amplitude of soil moisture. That meant the fluctuation

amplitude of soil moisture was positively correlated with soil porosity.

(ii) Plant interception and openings

Wang et al. (2010) identified that the plant interception yield was the amount of the interception of rainfall by the branches and withered leaves. But this value was difficult to obtain through experiments. Liu et al. (2015) pointed out that the maximum plant interception yield was scientific as the criterion for plant interception, which was positively correlated with rainfall intensity. She also pointed out that the relationship between leaf area and maximum leaf interception followed a linear function.

And also, it showed that the intercepted water was in a dynamic equilibrium state. It was consistent with the research findings from Liu et al. (2014).

The velocity of overflow was slightly lower than the one of runoff. It can be seen that root splitting fissures and unstable water storage spaces not only have water storage functions, but also have the function of reducing runoff velocity. It was consistent with the research findings from Zhang et al. (2019).

(iii) T_0

It was known that T_0 followed a rule of 65% WH + 30% BS ($\bar{I}_{min}$) > 95% FH ($\bar{I}_{min}$). It can be seen (Niu et al., 2021) that T_0 followed a rule of FH ($\bar{I}_{min}$) > BS ($\bar{I}_{min}$). Moreover, Wang et al. (2019) showed that the runoff starting time on a 4° bare soil slope was 4 minutes earlier than on a newly grown grass slope. It was known that the steeper the slope was, the

earlier the runoff starting time T_0 (Miao, 2020) was. Chang et al. (2019) and Wu et al. (2010) had similar conclusion, but their findings did not have quantitative data about withered herbs-covered slope. Based on the above conclusion, it can be concluded that $WH(\bar{I}_{min}) > FH(\bar{I}_{min}) > BS(\bar{I}_{min})$. It was consistent with the research findings from Chen et al. (2017), Christoph (2017), and Xiao et al. (2009).

The conclusion that $FH(\bar{I}_{min}) > FH(\bar{I}_{mid}) > FH(\bar{I}_{max})$ was consistent with the research findings from Feng et al. (2014) and Sheng et al. (2016). According to the above conclusions and Scenario 6 ($\bar{I}_{min}$) > Scenario 7 ($\bar{I}_{min}$), it was clear that $WH(\bar{I}_{mid}) > FH(\bar{I}_{min})$.

It was known (Wang et al., 2019) that T_0 of bare soil followed the law of $BS(\bar{I}_{min}) > BS(\bar{I}_{mid}) > BS(\bar{I}_{max})$. When the rainfall intensity increased, the larger the slope was, the greater difference the T_0 of bare soil had. Based on the above conclusions, it can be concluded that T_0 followed the rules of $WH(\bar{I}_{min}) > FH(\bar{I}_{min}) > BS(\bar{I}_{min}) > BS(\bar{I}_{mid}) > BS(\bar{I}_{max})$.

(iv) Q_0

From the direction of Q_0 before runoff producing, it can be seen that Q_0 includes three parts. They are plant interception, water storage from plant openings, and water storage from soil. The plant interception here is characterized by maximum plant interception yield. The water storage from plant openings is closely related to the space from fissures caused by root splitting and unstable water storage space. The yield of water storage from soil is closely related to soil depth.

(v) C_{ro} and C_{sy}

The research of C_{ro} showed that within 60 minutes of rainfall, there was a positive correlation between the growth of fresh herbs and rainfall intensity. As the duration of rainfall increased, there was a negative correlation. It inferred that the above situation was due to the increase in rainfall intensity over time, which increased the rate of rainwater infiltration into the soil, leading to a gradual decrease and stabilization of the increase in runoff coefficient.

The research of C_{sy} showed that the effects of runoff and sediment production followed the pattern of $BS > WH > FH$. During the rainfall process, the stems, leaves, and withered stems underwent their own elastic deformation. Combining with their rain resistance and water storage functions, they reduced the kinetic energy of rainwater and enhanced the erosion resistance of primary vegetation covering.

(2) Evaporation mechanism of primary vegetation covering

Charles (2017) pointed out that plants absorbed water through their roots and transpired excess water through stems and leaves. Gu et al. (2023) pointed out that the rate of water evaporation followed the trend of $BS > FH$. However, due to the rapid evaporation of water in bare soil, it was more prone to get cracks than that in fresh herbs. Mou et al. (2023) showed that as the crack width increased from 5 mm to 15 mm, the corresponding evaporation flux on the crack surface

increased from 18% to 51%. Preliminary analysis indicated that the stems and leaves blocked sunlight, regulating and creating a relatively stable microclimate environment to reduce evaporation of primary vegetation covering.

4. Conclusions

In-situ artificial simulation rainfall experiments were conducted to study the effects of bare soil and herbs on the infiltration, runoff, and sediment producing mechanism in Jinyang Ancient City.

(1) The rainfall of 13.4 mm was the threshold for changes in soil volume moisture content at depth in 0.5 m. When cumulative rainfall was no less than 13.4 mm, the volatility of soil moisture followed a trend of $0.5\text{ m} > 1\text{ m} > 1.5\text{ m}$. After rainfall, the soil at depths in 0.5 m and 1.5 m was in a rapid dehumidification stage within 1.5 hours and a stable dehumidification stage after 1.5 hours. The soil at depth in 1m was basically in a stable dehumidification stage. The fluctuation amplitude and the peak migration rate of soil moisture were positively correlated with soil porosity. The time of initial peak of soil surface volume moisture content was negatively correlated with its initial value before rainfall and rainfall intensity.

(2) The rainfall intercepted by plants was in a dynamic equilibrium state. Root splitting fissures and unstable water storage spaces not only had the function of water storage, but also had the one of reducing runoff velocity.

(3) For fresh herbs and withered herbs, Q_0 was negatively correlated with T_0 , and the interception effect of withered herbs was better than fresh herbs. T_0 followed the rules of $WH(\bar{I}_{min}) > FH(\bar{I}_{min}) > BS(\bar{I}_{min}) > BS(\bar{I}_{mid}) > BS(\bar{I}_{max})$, $WH(\bar{I}_{mid}) > FH(\bar{I}_{min}) > FH(\bar{I}_{mid}) > FH(\bar{I}_{max})$. Q_0 followed the rules of $WH(\bar{I}_{mid}) > FH(\bar{I}_{max}) > FH(\bar{I}_{mid}) > FH(\bar{I}_{min})$.

(4) The impact of rainfall intensity on the values of C_{ro} of fresh herbs decreased over time, and the impact of rainfall intensity on the values of C_{sy} of fresh herbs increased over time. The values of C_{ro-60} followed a rule of $WH(\bar{I}_{mid}) > FH(\bar{I}_{max}) > FH(\bar{I}_{mid}) > FH(\bar{I}_{min})$. The values of C_{ro-120} and C_{ro-180} followed a rule of $FH(\bar{I}_{min}) > FH(\bar{I}_{mid}) > FH(\bar{I}_{max})$. The values of C_{sy-60} , C_{sy-120} , and C_{sy-180} followed a rule of $FH(\bar{I}_{max}) > FH(\bar{I}_{mid}) > FH(\bar{I}_{min})$. The values of C_{sy-120} followed a rule of $WH(\bar{I}_{mid}) > FH(\bar{I}_{max})$.

(5) Fresh herbs and withered herbs enhanced the erosion resistance by intercepting and increasing the infiltration rate of rainfall and reducing the kinetic energy of rainwater. The stems and leaves blocked sunlight, regulating and creating a relatively stable microclimate environment to reduce evaporation of primary vegetation covering.

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